POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain different stages of computer evolution with reference of generation of computer.
   b) Explain instruction cycle state diagram with interrupt.
2. a) How overflow be detected in computer? Verify the operation (7) / (-3) using signed 2's complement method.
   b) Compare and Contrast between micro-program and hardwired control unit. Explain the micro program sequencer with example.
3. a) What is interrupt cycle? Explain how instruction be processed with interrupt in computer along with necessary diagram.
   b) What is m-way interleaving? Explain different types of memory interleaving.
4. Consider a system with main memory (MM) consisting of 8K blocks, a cache memory consisting of 256 blocks and a block size of 16 words- what will be the word field, block filed and Tag field length? How many bits are there in main memory address? If the system uses direct mapping and associative mapping techniques.
   b) What are the drawbacks of programmed I/O and interrupt driven I/O? Explain how DMA overcomes those drawbacks.
5. a) Assume that pipeline has K=8 segment and execute n=125 tasks in sequence. Let the time taken to process a sub-operation in each segment is 30 sec. Calculate the speed up ratio in the pipeline.
   b) What type of hardware and software performance issues are seen in multi-core computers? Explain.

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6. a) Explain different interconnection structures in multiprocessors.
   b) What is cache coherence problem? Explain MESI protocol for solving cache coherence problem.
7. Write short notes on: (Any two)
   a) RISC vs. CISC
   b) Hardware Description language (HDL)
   c) Logic Microoperation

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